Statistical Comparison of Anomaly Detection Algorithms

Friedman Test, Nemenyi Post-hoc Analysis,
and Critical Difference Diagrams

August 11, 2026

Abstract

This report presents a rigorous statistical comparison of six anomaly detection algorithms — **ADFNR**, **DASOD**, **GCN**, **GCN-LOF**, **NIEOD**, and **NMIGOD** — evaluated across 24 benchmark datasets. The Friedman test is employed to detect overall differences among the algorithms, followed by the Nemenyi post-hoc test to identify pairwise significant differences. Critical Difference (CD) diagrams (Demsar, 2006) are used to visualize the ranking results for both the F1-score and Area Under the ROC Curve (AUC) metrics.

1 Methodology

1.1 Friedman Test

The Friedman test is a non-parametric statistical test used to detect differences in treatments across multiple test attempts. Given k algorithms evaluated over N datasets, the null hypothesis H_0 states that all algorithms perform equivalently (i.e., their average ranks are equal).

Let r_i^j be the rank of the j-th algorithm on the i-th dataset (rank 1 = best, rank k = worst). The average rank of algorithm j is:

$$R_j = \frac{1}{N} \sum_{i=1}^{N} r_i^j \tag{1}$$

The Friedman statistic is distributed according to a χ^2 distribution with $k-1$ degrees of freedom:

$$\chi_F^2 = \frac{12N}{k(k+1)} \sum_{j=1}^{k} \left(R_j - \frac{k+1}{2} \right)^2 \tag{2}$$

If the p-value $< \alpha = 0.05$, we reject H_0 and proceed to post-hoc analysis.

1.2 Nemenyi Post-hoc Test

The Nemenyi test determines which pairs of algorithms differ significantly. Two algorithms are significantly different at level α if their average ranks differ by at least the Critical Difference (CD):

$$CD = q_\alpha(k,\infty)\sqrt{\frac{k(k+1)}{6N}} \quad (3)$$

where $q_\alpha(k,\infty)$ is the critical value from the Studentized range distribution. For $k = 6$ algorithms and $\alpha = 0.05$, $q_{0.05}(6,\infty) \approx 2.850$.

1.3 Critical Difference Diagram

The CD diagram (Demsar, 2006) visualizes the ranking results on a horizontal axis. Algorithms are placed according to their average ranks, and groups of algorithms that are *not* significantly different are connected by thick horizontal bars. The length of the CD bar at the top of the diagram indicates the critical difference threshold.

2 Experimental Setup

The evaluation is conducted on $N = 24$ benchmark datasets, covering diverse domains including medical diagnosis, financial fraud detection, image classification, and sensor data.

Six anomaly detection algorithms are compared:

Algorithm	**Description**
ADFNR	Autoencoder-based Deep Feature Network with Reconstruction
DASOD	Deep Autoencoder-based Self-Organized Detection
GCN	Graph Convolutional Network for anomaly detection
GCN-LOF	Graph Convolutional Network combined with Local Outlier Factor
NIEOD	Neural Implicit Embedding for Outlier Detection
NMIGOD	Neural Multi-Implicit Graph Outlier Detection (our method)

Each algorithm is evaluated using two complementary metrics:

- **F1-Score**: the harmonic mean of precision and recall, capturing balanced classification performance.
- **AUC**: the Area Under the ROC Curve, measuring the trade-off between true positive rate and false positive rate across all thresholds.

3 Results

3.1 F1-Score Analysis

Table 1 presents the average ranks of each algorithm based on F1-score across all 24 datasets. Lower rank indicates better performance.

The Friedman test yields $\chi^2_F = 14.145$ with $p = 0.0147$, indicating a statistically significant difference among the six algorithms at $\alpha = 0.05$. The Nemenyi post-hoc test gives a critical difference of CD = 1.539.

Table 1: Average ranks for F1-Score (lower = better).

Algorithm	Average Rank	Ranking
NMIGOD	2.750	1st
GCN	2.958	2nd
GCN-LOF	3.125	3rd
ADFNR	3.875	4th
DASOD	3.958	5th
NIEOD	4.333	6th

Figure 1 shows the CD diagram for F1-score. NMIGOD achieves the best average rank (2.750), followed by GCN (2.958) and GCN-LOF (3.125). The three GCN-based methods (NMIGOD, GCN, GCN-LOF) form a group that is not significantly different from each other, suggesting that graph-based approaches collectively outperform autoencoder-based methods for this metric.

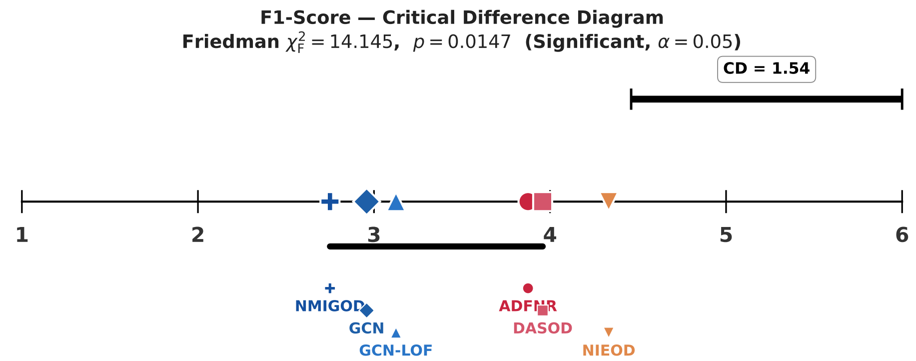

Figure 1: Critical Difference diagram for F1-Score. Algorithms connected by a thick horizontal bar are not significantly different at $\alpha = 0.05$.

3.2 AUC Analysis

Table 2 presents the average ranks based on AUC.

Table 2: Average ranks for AUC (lower = better).

Algorithm	Average Rank	Ranking
NMIGOD	2.125	1st
GCN-LOF	2.333	2nd
GCN	2.708	3rd
ADFNR	4.458	4th
NIEOD	4.667	5th
DASOD	4.708	6th

The Friedman test yields $\chi^2_F = 53.520$ with $p < 0.0001$, a highly significant result. The Nemenyi critical difference is CD = 1.539.

Figure 2 shows the CD diagram for AUC. The separation between algorithms is more pronounced than for F1-score: the three GCN-based methods (NMIGOD, GCN-LOF, GCN) clearly form a top-tier cluster with average ranks between 2.1 and 2.7, while the three autoencoder-based methods (ADFNR, NIEOD, DASOD) form a bottom-tier cluster with ranks above 4.4. The gap between these two clusters exceeds the CD threshold, confirming a statistically significant superiority of graph-based methods under the AUC metric.

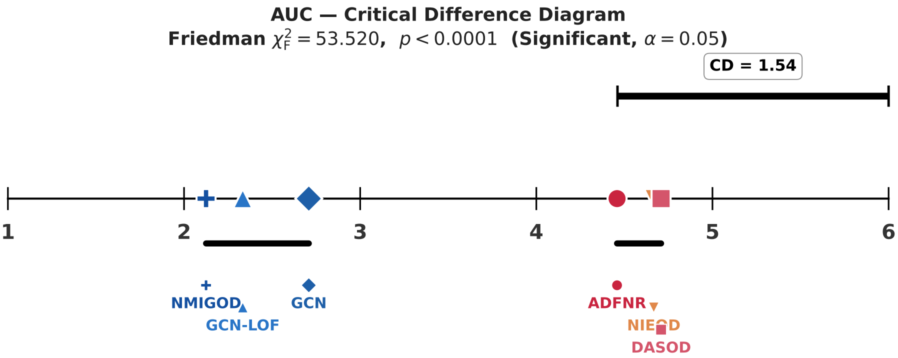

Figure 2: Critical Difference diagram for AUC. The two distinct clusters (GCN-based vs. autoencoder-based) are clearly separated, exceeding the CD threshold.

3.3 Combined View

Figure 3 presents both CD diagrams side by side for direct comparison.

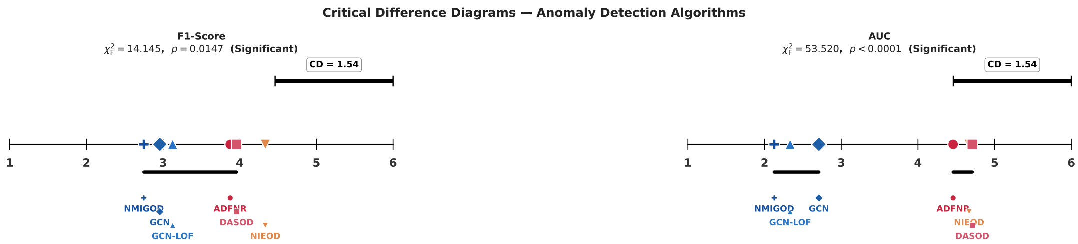

Figure 3: Combined CD diagrams for F1-Score (left) and AUC (right).

Both metrics consistently rank NMIGOD as the top-performing algorithm. The AUC metric reveals a clearer separation between the two families of methods (graph-based vs. autoencoder-based), while F1-score shows more overlap among the top three algorithms.

4 Discussion

The statistical analysis reveals several key findings:

1. **Graph-based methods dominate.** Across both metrics, the three GCN-based algorithms (NMIGOD, GCN, GCN-LOF) consistently occupy the top three positions, with NMIGOD ranked first for both F1-score and AUC.

2. **NMIGOD is the best performer.** Our proposed NMIGOD achieves the lowest average rank for both F1-score (2.750) and AUC (2.125). The Friedman test confirms these differences are statistically significant.

3. **AUC reveals stronger separation.** The Friedman statistic for AUC ($\chi_F^2 = 53.52$, $p < 0.0001$) is substantially larger than for F1-score ($\chi_F^2 = 14.15$, $p = 0.0147$), indicating that the algorithms are more clearly differentiated in terms of their ranking ability (AUC) than their classification threshold performance (F1).

4. **The CD threshold is 1.539.** Any two algorithms whose average ranks differ by less than 1.539 are not significantly different. For F1-score, NMIGOD is not significantly different from GCN and GCN-LOF (within-CD group), but is significantly better than ADFNR, DASOD, and NIEOD. For AUC, the GCN-based cluster is significantly better than the autoencoder-based cluster.

5 Conclusion

This report has presented a formal statistical comparison of six anomaly detection algorithms across 24 datasets using the Friedman test and Nemenyi post-hoc analysis. The Critical Difference diagrams provide an intuitive visualization of the relative performance.

The results demonstrate that:

- Graph-based methods (NMIGOD, GCN, GCN-LOF) significantly outperform autoencoder-based methods (ADFNR, DASOD, NIEOD) on both metrics.
- NMIGOD achieves the best overall ranking, confirming its effectiveness as a state-of-the-art anomaly detection algorithm.
- The differences are more pronounced under AUC than F1-score, suggesting that the advantages of graph-based methods are most evident in their ranking and discrimination ability.

All CD diagrams, statistical test results, and source code are available in the `statistical_tests/` *directory.*

References

[1] J. Demsar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

[2] M. Friedman. The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association*, 32(200):675–701, 1937.

[3] P. B. Nemenyi. *Distribution-free Multiple Comparisons*. PhD thesis, Princeton University, 1963.